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ORIGINAL ARTICLE

The importance of quality: How music festivals achieved commercial success

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Abstract This paper examines the characteristics that are important to current commercially viable massive music festivals when making decisions on who to hire while facing horizontal and vertical quality differentiation with heterogeneous preferences among intended consumers. A model of customer demand motivates the empirical analysis using a unique bundling problem in which the consumer faces an unknown element, depending on festival reputation for expected utility. The empirical analysis utilizes characteristics important to the negotiation between the festival and the band as input. Musical acts with an album recognized as being of a high quality are about 33 % more likely to be hired by a festival in the year of release and 50 % more likely the following year. A top touring band is 40 % more likely to be hired in the first year, but 45 % less likely to be hired in the following year, likely due to an increased cost of hiring. Festivals hire quality, unknown bands to take advantage of the lower costs of lesser known acts by leveraging a reputation for evaluating quality before the consumer. Other industries face similar input decisions, giving the results a broader application.

Keywords Input quality · Product characteristics · Music industry · Bundling · Expectations

1 Introduction

The USA has a rich history of major music festivals providing memorable performances to satisfied audiences. Productions such as the Newport Pop Festival, the Monterey Pop Festival, and Woodstock were early successes in terms of attracting huge audiences. Additionally, they are some of the most influential events

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in music history. They were not, however, commercial successes and often were not planned to make a profit. For example, the first major rock music festival in the USA, the Monterey Pop Festival, was operated as a nonprofit but ended operation $75,000 in debt (Brant 2008). In fact there were no festivals held annually in the same location on the scale of several current US festivals until the twenty-first century. Why was there such a failure to produce a product that currently exists in commercially successful forms? This paper explains the recent success in production of the annual music festival in terms of input substitution of the bands between established commercial successes and those bands of high quality without notoriety. Today, the largest festivals in the USA are Austin City Limits, Bonnaroo, Coachella, and Lollapalooza, all studied in this paper. For scale, Lollapalooza attracted 240,000 customers in 2010 and 270,000 in 2011, numbers which are similar for all festivals in this paper.[1]

Quality differentiation is usually discussed in terms of the difference in cost paid by the firm and the vertically differentiated levels of quality (Shaked and Sutton 1987; Dos Santos Ferreira and Thisse 1996). Music festivals must contend with some degree of vertical differentiation within each genre of music, as well as horizontal differentiation between genres. Festivals choose how to generate their product with the dual questions of what demand each additional act will provide weighed against the budget constraint of hiring in the context of their quality level. This is done by creating a "lineup," or compilation of musical acts that constitute a festival, each hired individually to a single festival contract. Within the lineup, there is a hierarchy of bands. The "headliners," or most highly demanded bands will receive the most prominent placement in promotional material, highest compensation, and are expected to draw the most customers.[2]

Among this hierarchy, there is considerable variation in genre of music, experience, and perceived quality. Quality here only refers to highly regarded contemporary contributions to the music industry. It is possible, for instance, for an artist to continue to profit off of a product of quality decades after its debut and without any additional works of significance in the interim. Quality measures in music must be somewhat subjective. Consumer preferences for music genres and bands are horizontally differentiated, with few absolutes in quality ranking (Peitz and Waelbroeck 2006). The expression "there is no accounting for taste" seems perfectly crafted to describe varied opinions about music. Any measure of overall quality of music must capture this heterogeneity in preferences.

The festival must make different decisions on who to hire for the headliners from those that will fill the smaller stages and less desirable times of the festival. I use a model of bilateral negotiation to explain the mutual hiring decision, estimating the probability of hiring with a conditional fixed effects logit. Because it is a negotiation, it does not depend solely on demand decisions. For that reason, a separate analysis measures the impact of various band characteristics on prominence

[1] http://www.chicagobusiness.com/article/20110730/ISSUE01/307309975/lollapalooza-vies-with-chicagos-top-conventions-in-spending-impact, Accessed: 1/27/14.

[2] Compensation is rarely made public, but conversations with music venue owners suggest the compensation structure varies considerably with demand. For an example of non-headliner fees see http://main-stageproductions.com/music, Accessed: 1/26/14.

within a festival. This model only includes bands that played a festival in a given year and determines the most important factors for a band's relative ranking in promotional material, using "rank" data estimated with least squares regression. Any differences in the demand results versus negotiation show where the festival must compromise between band characteristics they desire versus those they obtain in order to maximize profit.

A substantial literature is devoted to determining firm quality decisions and how they affect consumer choices. The early works of Lancaster (1971) and McFadden (1977) began the research into the importance of product characteristics. Wolinsky (1983) investigates the possibility that product quality can be used as a signal to consumers. Further empirical work has been done by Berry et al. (1995), Petrin (2002), and others that have explored empirical methods for estimating models of consumers with heterogeneous preferences and the impact of varied product characteristics. Mazzeo (2002), Chu (2010), and Matsa (2011) all explore the relationship between the level of competition and product quality.

Waldfogel (2012) shows that quality in the music industry has not declined with general revenue decreases in the past decade. He uses various "best of" lists of the top albums in specific time periods to measure quality. Using such widely read music review magazines as *NME*, *Spin*, *Mojo*, and others, I create a similar quality index. The reviewers are varied enough in intended consumers that any aggregate quality measures can range across a wide array of music preferences.

While consumers are usually assumed to know the characteristics of all of the bundled products, the theoretical model in this paper assumes there is an unknown element to the consumer in their bundle, a unique idea in the bundling literature. The model, in conjunction with the empirical evidence from a unique dataset, explains why quality and cost are not always perfectly correlated. This paper is the first to examine production of music festivals empirically, and results show that quality is important to music festivals. Nearly as important, however, is hiring inexperienced bands; explained by providing the same high level of quality without the corresponding high fees. The insights gained from the production of these music festivals contribute to the input substitution literature and can be extended to other industries facing similar negotiation decisions.

2 Background

The festivals mentioned in this paper have all experienced an increasing presence since inception. To be sure, there have been a number of other festivals of comparable size that failed to achieve success during the same period and are no longer in existence.[3] The continued presence of these productions shows an improved understanding by the festival of how to hire the appropriate lineup for

[3] For some other recent music festivals with continued success see Sasquatch!, Pitchfork, and Governors Ball music festivals. These festivals were not on the scale of those in this paper, and generally began more recently; they are now increasing in size very rapidly.

profitability. The hiring mix which these festivals use involves artists that vary widely in popularity and quality.

The four US festivals in this paper were chosen because of their massive size which allow for considerable artist substitution. Additionally, these festivals provide diversity of genres, consistent performances, and availability of data across years. There are, of course, many other festivals that take place in the USA, but none of the size and scope of these festivals during the period studied. Glastonbury, in the UK, was included for three reasons: the first being availability of additional data from a similar festival in an English-speaking market. The second is to determine whether the addition of one of the older, more successful massive festivals changed results. Results were not qualitatively changed by the addition, but the coefficients gained some significance.[4] The third is to allow for the inclusion of an English music publication, *NME*, in the quality measures.

The standard models of quality differentiation are usually written about in terms of a choice between high- and low-quality inputs and their difference in costs. The music festival faces consumers with broad horizontal differentiation in preferences, represented by genres of music, as described in Hotelling (1929). Each festival then contends with additional elements of vertical quality decisions determined in conjunction with these horizontal choices as explained in Shaked and Sutton (1987). Jacobson and Aaker (1987) examine quality choices in the context of competition, and Rhee (1996) models quality decisions in the face of unobservable consumer heterogeneity. Closer to this paper, Manova and Zhang (2012) use multi-product firms to establish that within firm quality differentiation is important in boosting total sales. Results in this paper show that levels of quality are differentiated within the music festival.

Any analysis of music festivals must extend beyond the traditional theory of differentiation. The festival industry may be a case where bands of the highest recent quality are not perfectly correlated with the highest fees for performance. Prior popularity plays an important role, allowing for persistent demand despite a lack of quality in the near past.[5] This industry must be considered with these caveats. Rob and Waldfogel (2006) find that ex-post valuations of albums often fall below ex-ante valuations of albums. They attribute this largely to simply growing tired of music over time. If the utility of an artist declines with further listening, recent quality will play an important role in attracting customers to see a live performance.

These decisions are also similar to that of a streaming video service, such as Netflix or Amazon Prime. These services charge a fixed monthly or annual fee for unlimited viewing from their library of film and television. The clear objective is to provide, in any month, sufficient entertainment value to the consumer to pay for the service. These services seem to attempt to achieve this utility without the availability of many recent "blockbusters," or commercial success of tremendous

[4] A model excluding Glastonbury is available upon request.

[5] Correlation coefficients between quality measures and commercial success show little relationship between the two. Alternatively, commercially successful albums and successful tours for band are highly correlated.

popularity.[6] The industry has a similar constraint of high licensing fees for the most demanded product, requiring a similar mix of several "headliners" and many obscure, but less expensive films.

The music festival is an example of a firm that takes advantage of product bundling.[7] As first noted in Stigler (1963) and extended in McAfee et al. (1989), under certain conditions a monopolist can maximize profits by exclusively selling bundles. The music festival is a pure bundle. It is impossible to watch every performance in a festival as there are always simultaneous performances.[8]

Each music festival uses an exclusivity clause to control the ability of participating bands to perform independently of the festival within close proximity to the festival (Kot 2010). This is a stringent exclusive dealing clause for bands hired, prohibiting musicians from playing again in the same region for several months around the festival dates. For this reason, some potential festival participants may choose to never accept a contract. Alternatively, some bands may find playing many festivals to be an optimal strategy (Hiller 2014). Combined with the somewhat transient nature of touring bands, the music festival has an effective monopoly on the local performance of the participating musicians.

This stands in stark contrast to the standard musical performance where a venue provides one or two primary bands with a considerably lower ticket price, acting much like the examples of pure bundling firms provided by Adams and Yellen (1976).[9] The festivals may have an advantage in information when creating their bundle. The idea of using informational leverage and quality bundling as a signal, similar to the evaluation by festivals in this paper, is put forward by Choi (2003) to explain how a firm may use a well-known high-quality product with a newly introduced product to encourage the new product's purchase. In the literature of media bundling, Shiller and Waldfogel (2011) look at various pricing possibilities of songs in the digital space and Crawford and Yurukoglu (2012) examine the potential effects of unbundling television channels.

Montoro-Pons and Cuadrado-García (2011) document the complementary nature of recorded music and live performances, and Mortimer et al. (2012) explain that concert revenue and the amount of time bands spend touring have increased in the period since file sharing began. Music festivals and touring more broadly have become a substantially bigger business in the recent past; corresponding with easier consumer access to new music, increasing demand for concerts, and increasing concert prices.

The festival receives revenue from the tickets sold to consumers as well as concessions and memorabilia available for sale within the festival grounds. These festivals use restricted entry and exit, making navigation to nearby dining alternatives difficult. Combined with the fact that the festival controls the level of

[6] For a discussion of the Netflix model see http://blogs.reuters.com/felix-salmon/2013/06/13/why-netflix-is-producing-original-content/, Accessed: 1/30/2014.

[7] Bundling in this paper refers only to musical acts, the primary draw of the festival. Some "VIP" packages bundle other services, such as food and drink, which will not affect hiring decisions.

[8] The organizer of Lollapalooza discussed the need of the festival to bring in varied types of artists in an article on MTV.com in 2012. http://www.mtv.com/news/articles/1691162/lollapalooza-2012-perry-farrell.jhtml, Accessed: 1/27/14.

[9] For a comparison of music festival versus local venue ticket prices (see Hiller 2014).

concession competition, there is potential for large profits from sales within the festival. Theory suggests that this could mean a lower ticket price than would be charged without concession profits, but the decision of which band to hire is still based on the draw of the artist versus their cost because internal festival sales are unlikely to be dependent on the band hired.

3 A model of festival demand

The ultimate objective of the festivals studied in this paper is profit maximization.[10] It is illustrative to simplify the operations of a festival to a basic level in order to consider the festival's decision-making process when hiring a band. A festival depends on an array of bands differing in popularity and genre to create its product. The headliners are well-known and most consumers will have a set expected utility for seeing them. Other bands will be placed in time slots when the headliners are not playing or on smaller stages, and the utility from seeing these acts is more variable. These bands are not prominent or highly demanded, either due to little success or recent entry into the market.

In the following model, I simplify this hierarchy of bands into two types, known and unknown. This is undoubtedly an oversimplification of the categories a festival uses to choose bands, which likely include separate groups for recent and past success as well as different categories for aspiring bands. This model would not benefit as an example from the added complexity; however, in the empirical analysis, these dimensions are captured in the variables describing the performers.

Unlike most bundling problems, the consumer does not know the quality of the unknown band's live performance and depends on the hiring reputation of the festival for expected utility. Festivals will hire one of each type based on characteristics of the band and the fee they must pay in order to hire them. Specifically, festival i books only two bands from the entire pool of potentials for its production in period t, with representative consumer l who is deciding whether she will purchase a ticket. Band k is known to the consumer, and band n is unknown. Under these circumstances, consumer l makes her decision expecting she will receive utility u_{likt} from the known band. Utility from the known band is increasing in the quality of band k, q_{kt}, in period t.

The consumer then expects to receive u_{lint} from the unknown band which is increasing in the past quality level of festival i, q_{it-1}. Ultimately, q_{it-1} is based on the reputation of the festival for hiring high-utility unknown bands in previous periods and is assumed to be equal to q_{nt-1} in this simple example. Implicit in this model is the assumption that the festival will not hire only headliners for the multi-day event, but optimizes profit by mixing known and unknown bands. The consumer then buys the ticket if:[11]

[10] There are possible alternative objectives, such as the early festival motivation of maximizing utility of consumers without generating a negative profit. However, given the nature of these festivals and the corporations that own them, profit maximization is not an unreasonable assumption.

[11] There are other potential reasons for attending a festival. Earl (2001) notes the seeming lack of logic in seeing a live performance when recorded copies are easily available. He explores the possibilities of the social side of concerts, release of inhibitions, and concert as ritual, among others.

$$u_{likt}(q_{kt}) + u_{lint}(q_{nt-1}) \geq p_{it}. \tag{1}$$

For two bands of equal quality, expected utility is assumed greater in viewing a known band. The festival must make its decision based on the desired level of quality for the known band, the level of quality it wants to establish or maintain for the next period with the unknown band, and the fees it must pay for each type of band.[12] The known band k is chosen from the set of known bands K, with a cost of Fee_{ikt}. The unknown band n is chosen from the set N, with a cost of Fee_{int}. Each fee is based on characteristics of the band, which will be discussed further in later sections. For now, it will suffice to say that the fees of each band are determined by their levels of quality, q_{kt} and q_{nt}. Fee_{kt} is assumed higher than Fee_{nt} for two bands of the same level of quality, which combined with the utility assumption are the primary reasons a festival has for hiring an unknown band. For any number of periods T, festival i then solves the optimization problem:

$$\begin{aligned} &\max_{q_{kt},q_{nt}} \sum_{t=1}^{T} p_{it} - \text{Fee}_{ikt}(q_{kt}) - \text{Fee}_{int}(q_{nt}) \\ &\text{s.t.} \;\; u_{likt}(q_{kt}) + u_{lint}(q_{nt-1}) = p_{it} \quad \forall t. \end{aligned} \tag{2}$$

The constraint for the festival assumes the price of the festival, p_{it}, is set based on the sum of consumer utility from known and unknown bands. In a single period model, the festival will attempt to set its price equal to the combined utility the consumer derives from the known band and expected utility based on the previous quality of the festival, and hire the unknown band that will accept the lowest fee possible. The reality of these festivals is different, as each decision must be made within the context of a repeated game of unknown length. Reputation becomes important in the repeated game, and the festival must consider how decisions will impact revenue in the next period.

Optimization allows the festival to set its price based on the expected utility of the consumers and maximize profit if they can reasonably predict the utility bands will provide. Being an accurate predictor of expected utility is the festival's advantage. The capacity of a festival is set prior to decisions on who to hire and determined by the limitations of the venue. Each festival in this study regularly sells out of tickets, so the model can be extended from a representative consumer to any number of consumers by assuming the festival attempts to set a price equal to the sum of consumer utility at its capacity.

No functional form is assumed for how the band's fee or consumer utility responds to quality. The assumptions used in order for this model to fit the observed hiring patterns of festivals are consumer utility increase at a similar rate in the quality of known and unknown bands, and fees increase more quickly in quality for the known band. This leads to festivals tending toward higher quality among the bands they hire that are unknown. Additional changes can be made to allow utility to vary by consumer; reputation can depend on more than merely the past period, and allowances can be made for varying types of bands beyond known and unknown.

[12] There may also be complementarities between the known and unknown bands. This model is intended to provide a simple illustration and could be expanded with considerably more complex relationships.

4 Empirical model specification

The primary empirical objective of this paper is to determine how music festivals make their production decisions, and using that knowledge to explain how firms with varying costs contend with quality. To do this, I first address what factors affect the likelihood of a band playing these music festivals and determine whether recent quality is an important variable in deriving these probabilities. If the model of known versus unknown bands is correct, newer high-quality bands should have a higher likelihood of participation. Second, I find what is important in assigning prominence within a festival among those bands that are hired to participate. Beyond the quality measures, I include various characteristics of bands that could plausibly affect the festival decision making.

4.1 Festivals' choices of who to hire

Two equations serve as the basis for the empirical study. The first is a profit function for any of the festivals in the sample, and the second a decision equation for each band. The unobserved reduced form expected marginal profit function for a festival hiring a band is:

$$\begin{aligned}\pi^*_{ijt} = {} & \text{Revenue}^*_{ijt}\left(\text{Experi}_{jt}; \text{Quality}_{jt}; \text{PastQuality}_{jt}; \text{Popularity}_{jt}; \text{PastPopul}_{jt}\right) \\ & - \text{Fee}^*_{ijt}\left(\text{Experi}_{jt}; \text{Quality}_{jt}; \text{PastQuality}_{jt}; \text{Popularity}_{jt}; \text{PastPopul}_{jt}\right) + \epsilon_{it}\end{aligned} \tag{3}$$

where the expected marginal profit is for festival i hiring band j in period t. This function requires assumptions that follow the structure of festival production. The firm creates the festival by determining the dates, procuring the space necessary, and then hiring the bands to fill the lineup, making the number of bands which can be hired exogenous from the decision of which bands are hired. The assumption implies that all festival costs are fixed and there are no marginal costs to hire a band except for the fee paid. Revenue and fees paid are dependent on the attributes of the band hired.

Marginal revenue is assumed to be linearly dependent on several band characteristics:

$$\begin{aligned}\text{Revenue}^*_{ijt} = {} & \gamma_1 \text{PriorFests}_{jt} + \gamma_2 \text{PriorFestRank}_{jt} + \gamma_3 \text{LastToured}_{jt} \\ & + \text{Quality}_{jt}\Theta + \text{Popularity}_{jt}\Gamma + \epsilon_{it}.\end{aligned} \tag{4}$$

The error term for the expected profit function is that of marginal revenue for the festival. PriorFests is a measure of the festival experience of a band in the last 2 years, used as a predictor of future demand. The festival is also likely to look at how the band was placed before, so PriorFestRank is the average previous ranking for a band if they played a festival within the last 2 years. The LastToured variable measures how much time has passed since the band has last toured. Quality is a vector of the various quality index variables used throughout the paper and their lag values, while popularity is a vector of the common measures of band popularity explained in the Sect. 5, as well as lag variables for each.

When producing a festival the bands are the inputs, and they must benefit to agree to participate. The band's profit function, also not observed, is:

$$\pi^*_{jit} = \text{Fee}^*_{ijt}\big(\text{Experience}_{jt}; \text{Quality}_{jt}; \text{PastQuality}_{jt}; \text{Popularity}_{jt}; \quad \text{PastPopularity}_{jt}\big) - \text{CostTouring}^*_{jit} - \text{OppCost}_{jit} + \epsilon_{jt}. \tag{5}$$

Band j is paid the fee negotiated with festival i in period t. The costs of participating are both explicit in terms of the expense of touring in that year for the band, and implicit in the opportunity cost of playing the festival. Implicit costs include foregone revenue from concerts in the surrounding region if not for the exclusive deal required by the festivals.

The opportunity costs could vary considerably depending on the band, so a band fixed effect will serve as the individual opportunity cost variable. Additionally, a year fixed effect measures whether the general opportunity cost changes over time, gauging the overall climate of the music touring industry. Touring decisions can vary by type of music played, prominence of the band, and whether the band is on hiatus or disbanded. Estimation without fixed effects rarely converged in tests in Sect. 6, likely indicating an incorrect specification.

In order for band j to be hired by festival i , two conditions must hold. First, the band must be more profitable than any band not chosen, $-j$:

$$\pi^*_{ijt} \geq \pi^*_{i-jt}. \tag{6}$$

Second, the benefit to band j must be greater than or equal to its alternatives, $\pi^*_{jit} \geq 0$. In order to gain tractability, I make a common economic assumption about the bands, that each will receive a fee equal to its explicit and implicit costs of playing, $\pi^*_{jit} = 0$. Equation 5 can now be substituted into Eq. 3, eliminating the fee paid to the band and providing an equilibrium if the hiring choice satisfies profit maximization:

$$\pi^*_{ijt} = \text{Revenue}_{ijt} - \text{CostTouring}_{jit} - \text{OppCost}_{jit} + \epsilon_{it} - \epsilon_{jt}. \tag{7}$$

The difference in profit of a band playing a festival versus those not hired is now:

$$\pi^*_{ijt} - \pi^*_{i-jt} = \text{Revenue}_{ijt} - \text{CostTouring}_{jit} - \text{OppCost}_{jit} - \text{Revenue}_{i-jt} + \text{CostTouring}_{-jit} + \text{OppCost}_{-jit} + \epsilon_{jt} - \epsilon_{-jt}. \tag{8}$$

The error term for the festival, ϵ_{it}, is differenced out for any year. With the assumption of a type 1 extreme value distribution for band error terms, this model can be estimated using a conditional fixed effects logit. The difference in profit between the chosen band and all others, $\pi^*_{ijt} - \pi^*_{i-jt} > 0$, is not observed. However, the dependent variable, whether or not a band played a festival in a given year, equals 1 for any band where this inequality holds and 0 otherwise.

4.2 Prominence within a festival

The capacity and stage layout of the festival is established before the lineup is determined. After all decisions on who to hire are made a festival must determine the

ordering of bands, establishing which bands will play on the larger stages and be advertised heavily to attract consumers. Scheduling conflicts and demands by some headliners will likely have to be sorted out in the negotiation process, but in general the festival makes these decisions based on many of the same characteristics used in the hiring decision with minimal input from most bands. The model testing the determinants of prominence within a festival is similar to the festival's revenue function:

$$\begin{aligned}\text{AveFestRank}_{jt} = {} & \gamma_1 \text{PriorFests}_{jt} + \gamma_2 \text{PriorFestRank}_{jt} \\ & + \gamma_3 \text{LastToured}_{jt} + \text{Quality}_{jt}\Theta + \text{Popularity}_{jt}\Gamma + \epsilon_{jt}.\end{aligned} \tag{9}$$

The sample for this model is limited to those bands which have been hired by a festival in a given year. The dependent variable for estimation, AveFestRank, is the average rank in band prominence of festival promotional material. No band fixed effects are included as the changing characteristics of bands from year to year should change the ranking, but time fixed effects are included. Additionally, there is no cost of touring included in this model as each band in this sample participates in a festival.[13]

5 Data

To test what determines a band's likelihood of playing festivals, I created a dataset of potential bands that could play these festivals each year. The bands which form this set are established by two sources. First, any band that has played in one of the five festivals examined is considered to be a potential performer. Additionally, each band which appears in one of the quality measures is included whether it has played a festival before or not. An overview of the variables used in this paper is available in Table 1.

Data on the festival lineups come from a variety of sources. Each of the observed festivals has a website with some archival history of past performances.[14] For most years of a festival's history, there are options to order artists by their expected demand, with headliners coming first and bands with lesser demand in descending order. Where this ordering is not possible, I accessed promotional posters from each year of the festival, noting prominence of name placement as a measure of expected demand. High-demand headliners are listed first and in a larger font, while a decreasing font and less prominent position are used as the relevance of the band decreases. The process of determining a ranking is slightly subjective, but general distinctions can be made between the various classes of bands as determined by the festival's expected demand.[15] Within the dataset, Coachella first appeared in 2001.

[13] All prominence models are estimated without a correction for selection bias, as the processes are assumed independent. As a robustness test, the models displayed in this paper were all run with a Heckman correction, with qualitatively similar results. These specifications are available upon request.

[14] Austin City limits: aclfestival.com, Bonnaroo: bonnaroo.com, Coachella: coachella.com, Lollapalooza: lollapalooza.com, Glastonbury: http://glastonburyfestivals.co.uk. All last Accessed: 10/11/2011.

[15] Each year's lineup for each festival was then manually checked against information on Songkick.com, a company which collects data on touring in the music industry.

Table 1 Variable list

Variable	Description
ACL	Austin City limits music festival, indicator (0/1)
Bonnaroo	Bonnaroo music festival, indicator (0/1)
Coach	Coachella music festival, indicator (0/1)
Lol	Lollapalooza music festival, indicator (0/1)
Glast	Glastonbury music festival, indicator (0/1)
Bestever	Indicator (0/1) for a music rating from Bestever.com
Mojo	Indicator (0/1) for a music rating from *Mojo*
Pitchfork	Indicator (0/1) for a music rating from Pitchfork.com
Spin	Indicator (0/1) for a music rating from *Spin*
NME	Indicator (0/1) for a music rating from *NME*
Metacritic	Indicator (0/1) for the metascore music rating from Metacritic.com
Fest	Indicator (0/1) for a band playing in any music festival in a year
Rating	Indicator (0/1) for a band receiving at least one quality rating in a year
TotalRating	The total number of quality measures a band is included in a year, discrete
AveRank	Average rank for quality indexes a band is included in a year, continuous
TopAlbum	Indicator (0/1) for a band with a top 200 gross revenue album in a year
TopTour	Indicator (0/1) for a band with a top 100 gross revenue tour in a year
PriorFests	The number of festivals a band has participated in its past, discrete
LastToured	How long ago a band last toured, discrete
FirstTour	Indicator (0/1) for a band producing its first tour in the sample
TourCosts	Indicator (0/1) for a band not touring outside of a festival in a year
FirstRating	Indicator (0/1) for a first quality rating by a band
EverFest	Indicator (0/1) if a band has ever played a festival before
PriorFestRank	The average previous ranking for a band if they played a festival within the last 2 years, continuous

Bonnaroo, Austin City Limits, and Glastonbury all first took place in 2002, and Lollapalooza became a permanent fixture in Chicago in 2005. All festivals were then held annually except Glastonbury, which was not produced in 2006.[16]

Quality measures are similar to those used in Waldfogel (2012), but are annual lists of the highest rated albums produced in the preceding year rather than a decade-long examination. These lists are produced by music themed magazines and websites, and represent a wide range of musical preferences.[17] In each of the lists, the top 30, 50, or 100 albums of the year are ranked by a quality measure such that $q_1 > \cdots > q_n$, where q_i is the quality of album i. All of the lists are from publications or websites produced in the USA or United Kingdom. The integer value of the ranking of an album and more importantly the band which produced the

[16] Glastonbury is a festival in the United Kingdom comparable in size, attendance, and hiring structure to the other four, included to increase the sample and improve estimates. Because it is outside of the USA all specifications were also run with Glastonbury excluded, and the results were not qualitatively different.

[17] The year-end lists are produced by BestEver.com, Metacritic.com, Pitchfork.com, *Mojo*, *NME*, and *Spin*.

album in each of the six publications is recorded.[18] Most bands do not appear in any of these rankings in a given year, meaning a zero is assigned to the band for this publication year.[19]

Preferences for music are horizontally differentiated. For all of the top album lists except for Metacritic and Besteveralbums, the editorial staff decides on their opinions of the quality of the year's production of albums and their relative rankings. Rankings vary across publications because of the varied preferences in music production. Total consensus of the highest quality music producers in a given year is an impossibility. Some heterogeneity in the rankings is crucial to examining how festivals make their decisions as consumers are similarly heterogeneous in preferences. The difference across the various measures of quality will be used to help determine which of the top album lists chosen have the biggest effect on festival hiring.

Metacritic creates a score based on a 100-point scale for albums released. They do so with a process that "curates a large group of the world's most respected critics, assigns scores to their reviews, and applies a weighted average to summarize the range of their opinions".[20] Different weights are assigned to different critics based on their perceived importance and stature, as determined by Metacritic, resulting in a weighted index of the best albums of the year as chosen by many critics, easily ranked by their numerical score.

Besteveralbums are different in that the retrospective rankings are not absolutely fixed at the end of the given year.[21] The firm allows users to submit their own list of the top albums and aggregates the results to create their list of the top 100 albums. Because of the possible fluidity of these rankings, the effect on festival hiring decisions may vary from the other quality measures. Specifically, it may be expected that as bands gain prominence their relative ranking on a changing list may rise, creating a positive bias on the relationship between these rankings and festival appearances. This bias should be less important in more recent years as there may not yet be the requisite time needed for any correction in popularity.

Some festival lineup decisions are likely driven by demand considerations other than quality. Album sales by a band are an obvious indicator of some degree of popularity. The "Billboard Top 200" is a list of the top 200 albums sold in a year, as determined by Nielsen Soundscan.[22] For each year, an indicator (TopAlbum) is applied to any band which reaches the top 200 in album sales. Additionally, the top touring bands may have an increased likelihood of being hired by festivals. Pollstar ranks the top 100 touring bands of the year on gross revenue. These are successful bands that can presumably demand a high fee for appearance in a festival. An

[18] I manually collected data on rankings from publication websites in order to ensure accuracy.

[19] All years from 2001 to 2010 are included for these lists of top albums, with the exception of Pitchfork in 2001.

[20] http://www.metacritic.com/about-metascores, Accessed: 10/11/2011.

[21] Top 100 Lists used from each year in the dataset were fixed when accessed on 7/14/2011.

[22] Bands included in the festival database were again manually cross-referenced against the Billboard lists available online. Soundscan uses point of service sales data in the USA, as well as digital sales for the years following the introduction of online retailers like iTunes.

Table 2 Summary statistics

Variable	Mean	SD	*N*
Fest	0.126		19,327
Rating	0.067		19,327
TopTour	0.014		19,327
FirstTour	0.072		19,327
TourCosts	0.555		19,327
FirstRating	0.022		19,327
EverFest	0.1981		19,327
TotalRating	0.092	0.494	19,327
AveRank	1.307	6.363	19,327
TopAlbum	0.023		19,327
PriorFests	0.38	0.749	19,327
LastToured	1.32	1.652	19,327
PriorFestRank	15.007	29.78	19,327

indicator (TopTour) is included for the list of the top 100 touring bands from 2002 to 2007.

To account for the length of time between tours, I have constructed a series determining the touring habits of bands outside of a festival in the given year, providing a variable (LastToured) to express the length of time since a band last toured. This variable is not entirely accurate for bands that toured before 2001. For that reason, I also include a variable for the first tour of a band in the sample (FirstTour) which is the first ever for any new band but only the first during the observed period for bands that toured before the beginning of the dataset. The variable TourCosts is an indicator that measures bands which have not toured in that year.[23] Construction of the touring variable required a threshold number of performances excluding festival appearances, set at five, to be considered as having toured in a given year.

5.1 Summary statistics

Information on the independent variables is available in Table 2 and observed festivals through the sample years in Table 3. The average number of total bands playing the five festivals each year from 2003 to 2011 is 555.3. The years 2001 and 2002 are excluded from the summary statistics to facilitate the creation of lag variables. This number increases steadily, but not monotonically throughout the time period with a minimum of 372 bands in 2004 and a maximum of 747 in 2011. In the years 2003, 2004, and 2006, only four of the five festivals are operational. The average number of bands per festival also increases substantially over the period. From 2003 to 2006, the average number of bands per festival is 95.78, and 2007–2011 that statistic is 134.9. These festivals expand in size as lineup construction is mastered and reputations are established.

[23] The touring data are again from Songkick. I was unable to locate a complete database of which bands toured by year. I inspected each band's tour dates by year.

Table 3 Number of performers in festivals

Year	ACL	Bon	Coach	Lol	Glast	Total	Mean (active)
2003	122	67	81	0	117	387	96.75
2004	98	77	85	0	112	372	93
2005	110	80	95	58	120	463	92.6
2006	115	86	95	107	0	403	100.75
2007	121	101	120	148	141	631	126.2
2008	126	114	133	118	148	639	127.8
2009	122	132	142	108	147	651	130.2
2010	121	152	145	127	160	705	141
2011	123	160	171	138	155	747	149.4

Table 4 Correlation matrix for inclusion in quality measures

	Bestever	Mojo	Pitchfork	Spin	NME	Metacritic
Bestever	1					
Mojo	0.26*	1				
Pitchfork	0.27*	0.21*	1			
Spin	0.33*	0.27*	0.34*	1		
NME	0.34*	0.36*	0.23*	0.33*	1	
Metacritic	0.23*	0.29*	0.32*	0.26*	0.196*	1
T25						
Bestever	1					
Mojo	0.34*	1				
Pitchfork	0.32*	0.19*	1			
Spin	0.37*	0.25*	0.35*	1		
NME	0.48*	0.41*	0.26*	0.36*	1	
Metacritic	0.26*	0.24*	0.32*	0.29*	0.23*	1

The first section includes all rankings. The second includes only the Top 25 albums of each year

* Significance at the 5 % level using Fisher's z test

Table 4 shows the correlation matrix for the six publications used to create the quality index. There is certainly overlap, but inclusion in one of the lists does not guarantee inclusion in any other. There are 1218 observations in which a band received inclusion in only one measure, 318 in two, 156 with inclusion in three, 82 in four, 42 in five, and 30 with inclusion in all six. The sample has been limited in the second part of Table 4 to only include the top 25 ranked albums of each year in each measure. The limitation is imposed in order to see how much inclusion consensus is achieved if all lists have an equal number of albums in each year. Consensus on top albums is difficult to come by among these measures, indicating coverage of many possible values over the horizontal quality dimension.

In any given year, an average of 2.3 % of bands will have an album in the Billboard Top 200, and 1.4 % will have a top 100 grossing tour. The average band

can expect to have an album included in at least one of the quality measures 6.7 % of the time, with a mean number of publications recognizing them of 1.64 for each album rated. Additionally, the average band will play a festival in 12.6 % of the years in the sample, with a mean of 1.26 festivals played in each of those years in which they participate. Bands in the sample are actively touring for 48.6 % of the available years.

6 Results

The model in Sect. 4 represents the negotiation process leading to decisions on who to hire between festival and band, and has an indicator for each band to determine whether they played a festival in a given year as its dependent variable. This model is then estimated with a conditional fixed effects logit. The marginal effects from the logit are available in Tables 5 and 6.[24] Quality is measured with a simple variable indicating whether a band is included in one of the lists (Rating) in a given year in the models in Sect. 6.1, and then measured by how many quality ratings (TotalRatings) a band is included in Sect. 6.2. As mentioned above, data on the top 100 tours are available for only 5 years in the sample, meaning the TopTour indicator is included as a robustness check where each model is tested with the reduced sample.

The first column of each table provides the results for a baseline model excluding the cost of touring (TourCosts), which the second column includes. The third column adds to the baseline with an indicator for the first time a band receives a rating (FirstRating) and whether they have ever played a festival before (EverFest).[25] The fourth column adds two interaction terms that help determine the importance of quality ratings in conjunction with other potentially relevant band characteristics (FrstRat × YrsToured, Rating × TopAlbum). The fifth column displays the results which include the indicator for a top touring band (TopTour), as well as its lag. All results referenced are from the model in column 4 when each provides similar results. I discuss any significant outliers by referencing which model they are in.

6.1 Models using a quality indicator

Table 5 provides estimates for the model using a simple measure of quality, whether a band is included in any of the quality measures. Inclusion in at least one quality measure (Rating) clearly increases the probability of playing a festival. The primary measure of popularity (TopAlbum) provides a similar effect. It is important to note the duration of the effect of each of these variables.[26] In the first year, a quality

[24] The raw results from each specification are not in the paper, but can be requested from the author.

[25] EverFest begins as a 0 value for each band, becoming a time-invariant constant only after a band plays their first festival in the sample. Because of the potential for an initial conditions problem alternative specifications, available by request, exclude the variable and depend on PriorFests to explain experience through a linear relationship.

[26] A lead variable was also tested for the rating variable, but never proved important. I also tested further lags for the TopAlbum variable, but none were significant.

Table 5 Hiring models with an indicator for quality

	(1) Fest	(2) Fest	(3) Fest	(4) Fest	(5) Fest
Rating	0.123***	0.110***	0.175***	0.171***	0.178*
	(0.04)	(0.03)	(0.05)	(0.05)	(0.07)
Rating ($t-1$)	0.248***	0.203***	0.251***	0.262***	0.293***
	(0.02)	(0.02)	(0.02)	(0.02)	(0.03)
Rating ($t-2$)	0.084***	0.094***	0.082**	0.084**	0.153***
	(0.02)	(0.02)	(0.03)	(0.03)	(0.04)
TopAlbum	0.241***	0.176***	0.259***	0.275***	0.314***
	(0.05)	(0.03)	(0.04)	(0.05)	(0.07)
TopAlbum ($t-1$)	0.031	0.032	0.054	0.063	0.229*
	(0.03)	(0.04)	(0.05)	(0.05)	(0.09)
PriorFests	−0.034***	−0.038***	0.062***	0.065***	0.002
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
PriorFestRank	−0.004***	−0.003***	−0.005***	−0.004***	−0.006***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
LastToured	−0.147***	−0.063***	−0.075***	−0.080***	−0.051*
	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)
FirstTour	−0.101***	−0.218***	−0.214***	−0.213***	−0.214***
	(0.01)	(0.02)	(0.02)	(0.02)	(0.03)
TourCosts		−0.493***	−0.494***	−0.501***	−0.542***
		(0.01)	(0.05)	(0.05)	(0.08)
FirstRating			−0.113**	−0.03	−0.024*
			(0.04)	(0.03)	(0.13)
EverFest			−0.388***	−0.392***	−0.476***
			(0.02)	(0.02)	(0.04)
FrstRat × YrsTour				0.046*	0.007
				(0.018)	(0.03)
Rating × TopAlbum				0.04	0.049
				(0.13)	(0.16)
TopTour					0.197*
					(0.1)
TopTour ($t-1$)					−0.222**
					(0.08)
Year FixedEffects	Yes	Yes	Yes	Yes	Yes
Band FixedEffects	Yes	Yes	Yes	Yes	Yes
Observations	19,327	19,327	19,327	19,327	5705
Pseudo (R^2)	0.195	0.214	0.246	0.248	0.316

Standard errors in parentheses

The dependent variable, Fest, is an indicator for inclusion in any festival. The marginal effect is interpreted as the percentage change effect on hiring probability derived from a one unit change in the variable. Because the models in columns 4 and 5 contain interaction terms, the marginal effect and standard errors are manually computed. Estimation is done by conditional fixed effects logit

* $p<0.05$; ** $p<0.01$; *** $p<0.001$

rating does not match the effect of a top album, but the coefficient on the first lag [Rating $(t-1)$] nearly meets that of TopAlbum alone. With the effects from each of the quality lag variables, the increased probability is sustained over several years and the combined impact is greater than that of a top album.

The model predicts that a touring band (TourCosts is 0) with median characteristics in terms of previous albums, experience, and previous ratings but no quality rating has a 50.9 % chance of participating in a festival in a given year when fixed effects are assumed to be zero. The prediction is limited to those bands on tour as not touring drives the probability to near zero. A band that receives a quality rating will see an increase in likelihood of playing a festival of about 17 % points, with lagging effects from the quality ratings in the future. All else constant this average band would successfully negotiate with a festival 67.9 % of the time, increasing the likelihood by about one-third. Production of a top 200 album has a definitive probability increase of approximately 25 % points in the same year, with minimal lagging effects.

While having a successful album is certainly more important than a quality measure in the first year, quality inclusion has a lasting and ultimately more substantial impact. The implication of these results is that festivals see popularity in an individual year from a top selling album as important in that year, but the quality measures are a better indication of continued success in the following years. To some extent, unproven quality can be substituted for commercial success in the hiring process.

The variables measuring band characteristics show that once quality and album sales are accounted for, relatively unknown bands are more likely to be hired for a festival. The EverFest variable shows that bands that have played a festival before are significantly less likely to be hired for a festival in the current year than those that have not, with an effect of approximately 39 % points. Although the relative unknown appears more likely to be hired, the first inclusion in a quality measure (FirstRating) either does not indicate unknown status or acts to counter that possibility. Indeed, the first inclusion in a quality measure may have a negative impact on that same probability. The likely reason for this negative effect is the timing of the quality ratings, which are released at the end of the year the festivals take place in. Subsequent years show a more informed effect.

If a band is known, festival experience, represented by PriorFests, increases the likelihood of being hired.[27] In the first two columns of Table 5, the number of prior festivals played in the past 2 years has a negative impact on festival probability. This is reversed in columns 3 and 4 as EverFest is included in the model, proving that among the bands that have played a festival, those that play more have an increased likelihood of being hired again. For those bands that have played festivals before having a lower rank is equivalent to a more prominent position in promotional material. The explanation for this difference comes from the known versus unknown model. Bands unknown to the consumer at the time of hiring will

[27] A referee raised the possibility of negative synergies among hired bands. An alternative model, available by request, uses the band's hiring results and prominence rank of each festival in the prior year rather than the total numbers, with qualitatively similar results.

accept a lower fee, and if chosen wisely will have a significant positive impact on the festival's reputation. Among bands known to consumers those with more experience and proven demand are much more likely to be chosen.

The band activity variables prove that although being unknown to final consumers increases the likelihood of being hired, being inactive decreases it. The estimates of the TourCost indicator show that all else equal, the touring costs of an inactive band decrease the likelihood of a band playing a festival by 50 %. Additionally, LastToured estimates show that each year since the last year a band toured decreases the likelihood of a band being hired by over 7 %. Of the two interaction variables included in column 4, only one is significant at the 5 % level.[28]

When accounting for bands in the top 100 in gross touring in column 5, there is evidence that the measure has some impact on who festivals hire. In the same year, bands with a top tour are more likely to play a festival but the next year sees a larger negative effect on that probability. Overall, the results are similar to the models excluding touring variables. Including a top tour indicator as a robustness check does not discount the effects of quality, and in fact may increase their magnitude.

6.2 Models using total inclusion in quality measures

Table 6 presents the marginal effects for models which use the total number of quality measures an album is included in, represented by the variable TotalRatings. The results are similar to the simple indicator model. Again, relative unknowns are more likely to be hired, likely due to the fact that they can be paid a lower fee. But as in the last section, bands that are hired by a festival in previous years are more likely to be hired again if they were prominently promoted by each festival they participated in. Additionally, estimates on the FirstTour variable show that there is a limit to the increase in probability of hire for an unknown. A band on its first national tour is significantly less likely to be included in a festival, with a decrease of about 21 %, all else equal. The fact that a band is touring for the first time in the sample makes it difficult for a festival to evaluate potential quality and fit for hiring.

The estimates on quality measures show a smaller increase in likelihood for inclusion in a single publication than was true in the previous model. For each additional publication an album is included in, there is a corresponding probability increase over a band with no albums of about 6 % points in the first year, 10 % points in the second year, and 4 % points in the third year, all statistically significant. These results indicate that although consensus is difficult to come by when measuring music quality, the more unanimous the high opinion of a band's quality the more likely a known band is to be hired. In fact, consensus among the quality ratings nearly guarantees participation in a festival if the band is interested. Having a top selling album provides an increase in likelihood which is similar to that in the previous model, with an effect of about 25 % points in the year of that album, diminishing rapidly in following years.

[28] Due to problems interpreting marginal effects with interaction terms, estimation with interaction terms is done by manually coding the derivative and using the delta method for standard errors.

Table 6 Hiring models with total quality inclusions

	(1) Fest	(2) Fest	(3) Fest	(4) Fest	(5) Fest
TotalRating	0.043***	0.052***	0.061***	0.060***	0.069***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
TotalRating ($t-1$)	0.078***	0.102***	0.140***	0.102***	0.140***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
TotalRating ($t-2$)	0.032***	0.046***	0.040***	0.040***	0.093***
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
AveRank	0.002***	0.002**	0.003**	0.003**	0.002
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
TopAlbum	0.232***	0.181***	0.260***	0.273***	0.305***
	(0.05)	(0.03)	(0.04)	(0.05)	(0.09)
TopAlbum ($t-1$)	0.032	0.037	0.056	0.062	0.214*
	(0.03)	(0.04)	(0.05)	(0.05)	(0.09)
PriorFests	−0.029***	−0.036***	0.057***	0.057***	−0.006
	(0.01)	(0.01)	(0.01)	(0.01)	(0.02)
PriorFestRank	−0.001***	−0.002***	−0.003***	−0.003***	−0.006***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
LastToured	−0.149***	−0.061***	−0.078***	−0.077***	−0.048*
	(0.01)	(0.02)	(0.01)	(0.01)	(0.02)
FirstTour	−0.102***	−0.224***	−0.215***	−0.213***	−0.224***
	(0.01)	(0.02)	(0.02)	(0.02)	(0.03)
TourCosts		−0.499***	−0.499***	−0.501***	−0.558***
		(0.01)	(0.05)	(0.05)	(0.07)
FirstRating			−0.107**	−0.177***	−0.081
			(0.04)	(0.05)	(0.10)
EverFest			−0.387***	−0.388***	−0.487***
			(0.02)	(0.02)	(0.04)
FrstRat × YrsTour				0.048*	.016
				(0.019)	(0.03)
Rating × TopAlbum				0.059	0.052
				(0.08)	(0.15)
TopTour					0.213*
					(0.09)
TopTour ($t-1$)					−0.222**
					(0.08)
Year FixedEffects	Yes	Yes	Yes	Yes	Yes
Band FixedEffects	Yes	Yes	Yes	Yes	Yes
Observations	19,327	19,327	19,327	19,327	5705
Pseudo (R^2)	0.194	0.212	0.246	0.246	0.318

Standard errors in parentheses

The dependent variable, Fest, is an indicator for inclusion in any festival. The marginal effect is interpreted as the percentage change effect on hiring probability derived from a one unit change in the variable. Because the models in columns 4 and 5 contain interaction terms, the marginal effect and standard errors are manually computed. Estimation is done by conditional fixed effects logit

* $p<0.05$; ** $p<0.01$; *** $p<0.001$

In column 5, the top tour is again important. In each model, having a top tour appears to mean a higher likelihood of being hired by a festival in the same year. The effect is slightly larger in the model using TotalRatings than in the simpler Rating model. Statistical significance is a question though, as the estimate never rises above a 5 % significance level and is insignificant in most models. Any positive effect is then negated by a considerable decrease in the same probability the next year, seen as the coefficient on TopTour ($t - 1$). The decrease is approximately 22 % points in each of the two models. This result seems counterintuitive on its face, as both the quality measures and top album lag estimates are positive. The touring variable is slightly different. It indicates a band having one of the 100 highest grossing tours. These bands are able to command high ticket prices and have little difficulty in generating revenue, and can demand a high fee because of outside options as a well-known band. The second year effect is likely not a lack of demand, but an inability to reach a mutually profitable agreement.

6.3 Prominence within a festival

To get insight into what band characteristics are most desired by a festival, I reduce the sample to those bands which play a festival in a given year and find what is important for prominence in the lineup. This sample is limited to those bands which were hired, leaving no problem of negotiation between festival and band. This model can be seen as a clearer look into how the festival anticipates demand, whereas the earlier models had to account for negotiations with and decisions by the bands as inputs. The dependent variable is the average festival rank, where a lower number means a more prominent position in the festival and all festival rank models are estimated with least squares regressions using robust standard errors.[29,30] Negative coefficient estimates then indicate that the given attribute increases a band's prominence or lineup "rank." Each of these tables contains only four columns as there is no consideration of the cost to the band of touring. The coefficients referenced are from column 3, except for references to TopTour, which are located in column 4.

Tables 7 and 8 provide the results for the model of Eq. 9 with quality measured by Rating and TotalRatings, respectively. Although quality increases the hiring probability in the same year as the ratings, these coefficients show that quality ratings have little to no impact on prominence. The timing of the ratings may play a role in this, as ratings are published at the end of each year and the festivals are all produced beforehand. The festival is then hiring these bands with some knowledge of their quality and expecting they will enhance the reputation of the festival in future periods, without much hope of the band increasing demand for the current period. The ensuing two periods show this to be true, as the estimates are significant and have a more substantial impact in both.

[29] This specification can lead to rank predictions that are negative, but the large sample size and the fact that this is not a prediction model should help to negate these concerns.

[30] Alternative ordered probit and log-linear models were tested, showed identical signs and relative magnitude, and are available by request.

Table 7 Festival rank models with an indicator for quality

	(1) AveFestRank	(2) AveFestRank	(3) AveFestRank	(4) AveFestRank
Rating	−2.477	−1.999	−2.255	−0.903
	(2.87)	(2.99)	(3.14)	(4.04)
Rating ($t-1$)	−8.215***	−9.010***	−8.986***	−7.894***
	(1.43)	(1.43)	(1.43)	(1.78)
Rating ($t-2$)	−6.185***	−6.606***	−6.588***	−7.869***
	(1.67)	(1.69)	(1.69)	(2.18)
AveRank	−0.0522	−0.0993	−0.0940	−0.0688
	(0.09)	(0.09)	(0.09)	(0.12)
TopAlbum	−19.51***	−19.45***	−19.96***	−12.95**
	(2.81)	(2.78)	(3.56)	(4.89)
TopAlbum ($t-1$)	−19.53***	−19.33***	−19.22***	−8.865*
	(3.04)	(3.01)	(3.05)	(4.48)
PriorFests	−9.464***	−5.765***	−5.765***	−6.495***
	(0.66)	(0.77)	(0.77)	(1.21)
AveFestRank	0.161***	0.140***	0.140***	0.188***
	(0.02)	(0.02)	(0.02)	(0.04)
LastToured	5.238***	5.144***	5.142***	7.062***
	(0.54)	(0.53)	(0.53)	(0.57)
FirstTour	7.595***	6.948***	6.890***	10.20***
	(1.74)	(1.73)	(1.74)	(1.96)
FirstRating		0.787	1.639	−1.413
		(2.69)	(4.39)	(5.71)
EverFest		−13.06***	−13.07***	−6.072**
		(1.42)	(1.43)	(2.16)
FrstRat × YrsTour			−0.274	0.683
			(1.20)	(1.72)
Rating × TopAlbum			1.130	−1.524
			(5.14)	(7.04)
TopTour				−19.64***
				(4.59)
TopTour ($t-1$)				−17.88**
				(5.74)
Constant	78.56***	82.19***	82.18***	64.99***
	(1.34)	(1.39)	(1.39)	(1.46)
Observations	3562	3562	3562	1656
(R^2)	0.3247	0.3516	0.3517	0.3723

Standard errors in parentheses

The dependent variable, AveFestRank, is the average "rank" of all the festivals band is in. Estimation is done by ordinary least squares regression

* $p<0.05$; ** $p<0.01$; *** $p<0.001$

Table 8 Festival rank models with total ratings measuring quality

	(1) AveFestRank	(2) AveFestRank	(3) AveFestRank	(4) AveFestRank
TotalRating	−0.531	−0.448	−0.463	−0.800
	(0.74)	(0.74)	(0.76)	(0.93)
TotalRating ($t-1$)	−3.330***	−3.673***	−3.674***	−3.440***
	(0.56)	(0.55)	(0.55)	(0.68)
TotalRating ($t-2$)	−2.103**	−2.197**	−2.196**	−2.818***
	(0.68)	(0.68)	(0.68)	(0.80)
AveRank	−0.0859	−0.135*	−0.134*	−0.0635
	(0.06)	(0.07)	(0.07)	(0.08)
TopAlbum	−19.93***	−19.88***	−19.90***	−12.76**
	(2.81)	(2.78)	(3.56)	(4.88)
TopAlbum ($t-1$)	−19.91***	−19.72***	−19.72***	−8.029
	(3.03)	(3.00)	(3.05)	(4.50)
PriorFests	−9.411***	−5.658***	−5.655***	−6.316***
	(0.66)	(0.77)	(0.77)	(1.23)
PriorFestRank	0.158***	0.136***	0.137***	0.186***
	(0.02)	(0.02)	(0.02)	(0.04)
LastToured	5.322***	5.236***	5.236***	7.156***
	(0.54)	(0.53)	(0.53)	(0.57)
FirstTour	7.939***	7.286***	7.239***	10.51***
	(1.74)	(1.72)	(1.74)	(1.95)
FirstRating		1.635	2.407	0.121
		(2.57)	(4.30)	(5.55)
EverFest		−13.13***	−13.14***	−6.030**
		(1.42)	(1.43)	(2.16)
FrstRat × YrsTour			−0.272	0.448
			(1.21)	(1.72)
Rating × TopAlbum			−0.0112	−1.958
			(5.02)	(6.88)
TopTour				−20.01***
				(4.58)
TopTour ($t-1$)				−17.34**
				(5.73)
Constant	78.41***	82.02***	82.02***	64.65***
	(1.34)	(1.38)	(1.39)	(1.44)
Observations	3562	3562	3562	1656
(R^2)	0.3259	0.3527	0.3528	0.3772

Standard errors in parentheses

The dependent variable, AveFestRank, is the average "rank" of all the festivals band is in. Estimation is done by ordinary least squares regression

* $p<0.05$; ** $p<0.01$; *** $p<0.001$

Results in Tables 7 and 8 make it clear that inclusion in additional publications is not as important as it was for hiring probability. The first lag in each model, the most important period, shows an estimate of rank increase of about 9 using Rating, with the corresponding TotalRating coefficient at 3.7 in the same period in Table 8. Confirming the lesser importance of quality ratings are the top album indicators. Without an album of unanimous quality included in each of the lists, the combined effects of 3 years will not match the single year rank increase of almost 20 places that comes from having a top album. This impact is almost as large in the lagged year, leading to the conclusion that festivals are hiring bands respected in quality measures for the effect on reputation in ensuing periods, and hiring bands with well-selling albums for their immediate impact on demand.

The effects of some simple band characteristics on determining prominence have reversed from their effect on probability of hiring. EverFest has a substantially negative impact on being hired in a given year. However, once hired the experience of having played before means a more prominent position in the lineup by over thirteen places in both models. This is further reinforced with PriorFests, where there is an effect of between 5.6 and 9.4 in improved rank in each model for each prior festival played. Confirming the importance of experience is that unknown bands on their first tour are likely to be lower in prominence by 7–10 places in the lineup. Additionally, for each year elapsed since a band has toured there is a drop in rank of close to 5.5. Opposite to hiring probability, experience is important to how prominently a band is placed and therefore to its expected effect on demand for the festival.

Including top 100 tours in the models makes it clear that they do not greatly affect the quality measure coefficients. What does change considerably is the estimate on having a top album and its lag. Much of the prominence effect of having a top album is eliminated as another demand variable is included. In fact, although TopTour and its lag were not important in hiring probability, they are now the single most important effect on rank within a festival with an increase in rank of 20 in the first year and 17 in the second. Festivals are cautious about hiring bands with commercial success, but place those they do in the most prominent positions.

Band experience still has an important effect on prominence in the model including top tours; however, the coefficient on EverFest is now less important than it was in previous models. The number of prior festivals is also lower, indicating that experience alone is not sufficient for significant promotion. An unknown band can still expect to be ranked lower. A band's first tour now means an even less prime position in the festival, correlated with a decrease in rank of 10.5 compared to 7 previously. Adding touring as a robustness check is more important in the prominence model than it was in hiring. The quality measures are largely unchanged, but much of the effect from having a top album is now transferred to a top tour. The experience of bands is shown to be important, but not as meaningful without quality and demand.

Comparing predicted rankings to actual rankings, the model tends to slightly underestimate prominence, where predictions for bands with a quality rating are underestimated more than those without. Analysis of the 50 most underrated bands reveals some interesting results. For underrated bands, 20 % were on their first tour

in the dataset. This includes reunion tours by Rage Against the Machine in 2007, Jesus and Mary Chain in 2007, Pavement in 2010, Soundgarden in 2010, and The Cars in 2011. Having no previous activity in the dataset, these popular bands were bound to be underrated. Of these 50, only two bands had released an album with a quality rating in that year, and one band did so in the prior 2 years. Only two had played a festival before in the dataset. Additionally, only one had a top selling album in the previous 2 years. These bands, filled with big names, are very prominent in a festival because of past success and quality coupled with an absence of similar events in recent years.

As a final prominence robustness test, I consider the possibility that the size of the festivals affects ranking. The general expansion of each of the festivals over time causes more slots to open and increases the average ranking of a festival, potentially biasing the rank results. In models excluded for space, regressions are tested with the rank of bands playing a festival as a percentage of the total spots available, where a lower percentage indicates a higher prominence.[31] The results reinforce the ranking models. Every sign remains the same and the coefficient on the percentage changes are slightly stronger than the raw results evaluated at their average. The varying number of performers in festivals does not bias the results.

7 Conclusion

Music festivals must carefully consider which bands to hire for their annual production. The period studied coincides with a time in which bands began to generate a larger percentage of their income from touring. The festival cannot then simply depend on hiring the most popular group of each genre and creating the appropriate demand. The difficulty of the task can be seen by the many large music festivals that have failed for lack of profitability.[32] In order to maintain the optimal lineup, these festivals must be able to minimize the costs of the headliners that will drive much of their demand, as well as recognize quality of unknown bands ahead of the wider base of music customers.

The festival hires unknown bands which its consumers will enjoy but are not yet familiar with. These bands benefit a festival because they can enhance reputation and customer experience at a lower fee. The magnitude of this effect is best considered with a band that has played a festival before, where their probability of being hired in a subsequent year is decreased by about 75 %. This is by far the largest negative effect on the likelihood of a touring band being hired. This is not to say newly formed bands are likely to be hired, the festival needs the opportunity to evaluate potential. However, bands without festival experience are more often hired. The most important measurable characteristic of these bands is their quality, specifically inclusion in quality measures. Inclusion in quality measures greatly increases the probability of hiring, where wide acclaim nearly guarantees festival participation in ensuing years. Commercial success, represented by a top album,

[31] These models are available by request.

[32] For examples see Vegoose, Langerado, Monolith, and All Points West music festivals.

increases the probability of hiring substantially in the first year as well. In fact, the impact is greater than a quality album. In subsequent years, however, this effect is minimal while the quality effect strengthens.

Once hired the important characteristics for prominence within a festival change, indicating that the festival is compromising on who to hire to produce the festival based on more than immediate demand considerations. Promotional decisions are based on expected demand of the bands hired and reaching the critical demand necessary for that year's production. In promotion, quality becomes less important and the demand characteristics of top albums and tours become more significant. The prominence models show these characteristics are important to festivals, but the higher fees necessary to hire the more successful mean they must substitute some quality bands with low popularity for established bands that could create the largest immediate demand.

The hiring practices of music festivals illustrate the ability of a firm to bundle various levels of quality in order to create a profitable venture. Results show that if the firm is able to identify characteristics of its inputs that will create future demand for the consumer but have not yet been recognized for their full value, then they may be employed for less than their quality level would warrant. The bundling of these inputs with known and unknown levels of quality creates the demand that a music festival needs to remain profitable. Higher quality allows bands to overcome the lower probability of experienced bands being hired that comes with having to pay higher fees.

This paper provides insight into the music festival, but further research may extend the idea to other industries and operations. Examples include industries where quality and cost are variable, but may not be perfectly correlated due to a lack of knowledge about the product. The most obvious industries are streaming online video services, movie production, and sports franchises.

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